

POD_EDGE_mono_DSP vs Pod_MultiFX_Chain

Firmware-structure visualization focused on the control responsiveness fix: which work runs in the audio callback, which work runs in digital polling, and how parameter changes reach DSP without sluggish or jumpy behavior.

Report

FigJam

EDGE Graphify

MultiFX Graphify

BEST REFERENCE

POD_EDGE_mono_DSP

Most explicit responsiveness repair: timer ISR plus audio-rate analog plus double-buffer handoff.

CLEAN COMPACT REFERENCE

Pod_MultiFX_Chain

Small, readable BUG-005 implementation with page/layer stored values and smooth reattach.

GRAPHIFY SCALE

1457 vs 60 nodes

EDGE is broad and dense; MultiFX is compact enough to inspect as a whole graph.

MOST USEFUL VISUAL STACK

Mermaid + GoJS

Text-native precision for docs plus interactive reading for structure comparison.

POD_EDGE_mono_DSP

high-value reference

Input surface

Daisy Pod controls plus external I2C OLED, external encoder, BAK, and CON buttons.

digital path

1 kHz timer ISR

Runs Pod digital control processing and external encoder debounce at a stable rate.

analog path

Audio callback analog read

Runs Pod analog processing at callback rate, matching the Daisy control timing expectation.

handoff

Double-buffered FxParams

Main loop writes the inactive parameter copy, uses a barrier, then swaps the active index read by the ISR.

DSP

Mono DSP chain

Input conditioning, drive, SDRAM delay, diffusion, tone shaping, limiter, and dual-mono output.

Both projects split analog and digital control timing. EDGE goes further because the external encoder and button surface need deterministic debounce independent of display and UI work.

EDGE stops relying on a variable main loop for external encoder debounce.

EDGE keeps I2C, GPIO reads, allocation, and print work out of the audio callback.

EDGE sends a coherent parameter snapshot into DSP rather than letting the ISR read a structure being edited.

MultiFX solves page switching jumps with stored values, movement detection, and slew.

Pod_MultiFX_Chain

compact reference

Input surface

Daisy Pod knobs, encoder, SW1, and SW2. No external OLED or external encoder stack.

analog path

Audio callback analog read

Runs ``hw.ProcessAnalogControls()`` in the callback following BUG-005 guidance.

page comfort

Stored positions and reattach

Each page and shift layer stores values; knobs reattach after real movement and then slew toward the physical target.

digital path

Main loop digital events

Encoder and buttons update page selection, shift, bypass, and waveform state in a 1 ms loop.

DSP

Serial effect chain

LFO modulation, Tube, Delay, and Reverb run as a compact multi-effect chain.